Data Analysis and Coding

Team Project 1

Asking your own questions of real survey data

The project

Your team chooses one of three real survey datasets, decides what is worth asking of it, and answers its own questions with the statistical methods covered in this course.

Step 1 — Choose a dataset

All three are downloadable from e-campus, each with a codebook and a starter notebook. Each holds 1,868 respondents, so the choice is purely one of subject matter.

All four of your hypotheses come from the one file you choose, so read its codebook first, and check that it can support all four tests below. The codebook lists every variable, its valid n and its value labels.

- `gss2024.csv` — **General Social Survey 2024.** Attitudes, behavior and demographics across American life, including media use for political news, confidence in the press, education, religion, party, ideology, happiness and work.
- `pew2025.csv` — **Pew American Trends Panel, Wave 165 (2025).** News and trust. How closely people follow news at four levels, trust in several kinds of source, how the news makes them feel, and whether they use and trust eight named outlets.
- `kpf2025.csv` — **Korea Press Foundation Media Users Survey 2025.** Media use in Korea. Days per week on thirteen media (eleven with a parallel news version), minutes per day for newspapers, TV and radio, and detailed evaluations of the press.

The files share the same missing-value codes, though not every file uses all three.

- **998** for "don't know"
- **999** for "refused or no answer"
- **−1** for "not applicable"

These are ordinary numbers in the file, so they must be recoded as missing before you calculate anything. The starter notebook loads the file and shows you how.

Step 2 — Write four hypotheses

Three are fixed by method so that everyone practices the same tools. The fourth is yours. Where you have no directional expectation, write a research question instead and say so.

Before you write any hypothesis, agree on one topic, a single question about the media that your team wants to answer, such as who trusts the news or how young people get their news. All four hypotheses should be about that topic, because four findings on unrelated topics do not add up to a discussion.

The simplest way is to keep one variable in all four, for example trust in the news, and pair it with a different variable each time. For the chi-square test, group it into categories such as high and low. If a required method does not fit your topic well, choose the closest hypothesis you can and say so in your report.

- **H1 — an independent-samples t test.** One variable that splits people into two groups, and one continuous outcome.
  - Let Levene's test choose between the pooled and the Welch version.
  - Report Cohen's d.
- **H2 — a chi-square test of independence.** Two categorical variables.
  - Every cell needs an expected count of at least 5.
  - Report Cramér's V.
- **H3 — a one-way ANOVA.** One variable with three or more groups, and one continuous outcome.
  - Check equal variances with Levene's test.
  - Follow a significant F with Tukey's HSD.
  - Report eta squared.
- **H4 — your choice.** A paired-samples t test, a one-sample t test, or a two-way ANOVA with its interaction.
  - A paired test needs two questions answered by the same people on the same scale.
  - A one-sample test needs a meaningful test value, such as a scale's midpoint.
  - Report Cohen's d (the mean difference divided by its standard deviation), or partial eta squared for a two-way ANOVA.

A scale with five or more ordered points may be treated as continuous. One with three or four points is safer treated as categorical. If a category holds very few people, combine it with a similar one or leave it out.

Keep it short

A rationale of two or three sentences for each hypothesis. That is all. This is not a literature review, and length earns nothing. Still, search the academic literature first, because it is better to base each hypothesis on academic sources and cite them.

A hypothesis is a statement about how the world works that could turn out to be wrong. It is not a pair of variable names.

- **Not a hypothesis:** *"We will examine the relationship between education and news use."*
- **A hypothesis:** *"People with more education follow the news more closely, because education builds interest in public affairs."*

Step 3 — The report

- **Introduction.** What you set out to find out, and why it is worth knowing.
- **Data.** The dataset, the variables, how you handled the missing codes, and any categories you combined or left out.
- **Hypotheses.** Your four, each with its rationale.
- **Results.** One section per hypothesis, in order, each with the Python code, its output and your own interpretation, set out in the same format as the lecture notes.
  - Write the numbers into sentences, including how many people each test used.
  - Report a hypothesis that is not supported as plainly as one that is.
- **Discussion.** What the results mean together, and what the analysis cannot tell you.

Submission

Put the whole report into a single PDF file and submit only that file.